

Shubham Kumar Singh

Software Engineer — Backend Systems & Applied AI | Java, Spring Boot, Python

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SUMMARY

Software Engineer with backend and full-stack internship experience, focused on Java/Spring Boot systems and applied AI. Built an end-to-end fraud detection and investigation platform spanning Java, Spring Boot, FastAPI, XGBoost, and PostgreSQL. Solved 200+ DSA problems with strong fundamentals in OOP, DBMS, Operating Systems, and System Design.

TECHNICAL SKILLS

Languages: Java, Python, JavaScript, SQL, C

Backend: Spring Boot, Spring MVC, REST APIs, Node.js, Express.js, FastAPI, JWT, RBAC, WebSockets

Databases: PostgreSQL, MySQL, MongoDB, Redis, pgvector

Frontend: React.js, TypeScript, HTML5, CSS3, Tailwind CSS

AI/ML: XGBoost, OpenAI API, LangChain, RAG, Prompt Engineering, Gemini API, Groq API

Tools & Practices: Docker, Git, GitHub, Postman, API Design, Unit Testing, Load Testing, Agile

Core CS: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks, System Design

EXPERIENCE

Detagenix

Full Stack Developer Intern

Indore

Oct 2025 – Feb 2026

- Led planning and delivery across 3+ full-stack projects, serving as Lead Developer on 2, owning architecture and implementation in an early-stage startup environment.
- Built and optimized 10+ REST APIs with JWT authentication and RBAC, improving average API response time by 30% through query optimization.
- Developed 45+ reusable React components, improving cross-device UX and reducing page-load overhead across application interfaces.
- Resolved 30+ cross-stack bugs across concurrent projects using Git, Postman, and Agile development practices.
- Integrated an AI-powered chatbot to automate query handling and improve user interaction with application services.

PROJECTS

AI-Powered UPI Fraud Detection & Investigation Platform

Java 21, Spring Boot, Python, FastAPI, XGBoost, PostgreSQL, pgvector, React, TypeScript, Docker

- Engineered an end-to-end fraud detection and investigation platform on a synthetic dataset of 10,732 UPI transactions, modeling 6 fraud scenarios (account takeover, new device, unusual amount/location, new beneficiary, transaction bursts).
- Trained an XGBoost classification model achieving 0.9907 ROC-AUC, engineering behavioral features (transaction velocity, amount deviation, device/location change, beneficiary history) for explainable risk scoring.
- Built a FastAPI ML inference microservice decoupled from a Java 21/Spring Boot backend, implementing a polyglot service architecture separating model inference from transaction and case-management workflows.
- Designed a PostgreSQL + pgvector data layer powering RAG-based retrieval of semantically similar historical fraud cases for investigator-facing risk analysis and case triage.
- Integrated Redis caching for frequently accessed behavioral and transaction-risk data, reducing repeated database computation during fraud analysis.
- Built a 3-stage investigation workflow (OPEN → UNDER_REVIEW → RESOLVED) exposed through a React + TypeScript dashboard; containerized the full stack with Docker for reproducible deployment.

AI-Powered Interview Preparation Platform

React.js, Node.js, Express.js, MongoDB, OpenAI API, Web Speech API, Docker

- Built a full-stack AI interview platform generating role-specific, difficulty-adaptive questions via the OpenAI API, with an adaptive MCQ engine that adjusts difficulty based on live performance.
- Implemented speech-based interview interaction using the Web Speech API for spoken-answer transcription and AI-assisted evaluation, plus an AI resume analyzer that scores candidate skills against job requirements.
- Architected a Node.js/Express + MongoDB backend into modular services (question generation, evaluation, resume analysis), isolating AI-intensive operations from standard CRUD APIs; Dockerized for reproducible deployment.
- Built a real-time evaluation and feedback loop connecting candidate responses to the AI scoring engine, delivering end-to-end interview simulation with minimal latency between answer submission and feedback.

EDUCATION

Gyan Ganga Institute of Technology and Sciences, Jabalpur

B.Tech, Computer Science and Business Systems — CGPA: 7.5/10

2022 – 2026

ACHIEVEMENTS

- Solved 200+ Data Structures & Algorithms problems across LeetCode and GeeksforGeeks.
- Competed in Smart India Hackathon 2023 and 2024.
- Competed in TCS CodeVita Season 12 and 13.